

Harshit Chaudhary

Junior Undergraduate, Department of Mechanical Engineering
Indian Institute of Technology Kanpur

✉ harshitc21@iitk.ac.in | ☎ +91-9528733787
🔗 [HarshitC-27](#) | [in](#) Harshit Chaudhary

ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	7.5/10
2021	Class XII (ISC)	St. Conrad's Inter College, Agra	96.3%
2019	Class X (ICSE)	St. Conrad's Inter College, Agra	96%

ACHIEVEMENTS

- Secured **1st place** for the Ideation Document in **Robofest Gujarat 3.0**, thus receiving an initial grant of **Rs.50,000** *Jun'23*
- Bronze Medal, Inter - IIT Tech Meet 11.0** - Pluto Drone Swarm Challenge in a team of 10 shortlisted people *Feb'23*

WORK EXPERIENCE

SURGE 2023 | Research Intern | *Mentor: Prof. Tushar Sandhan* *May'23-Jul'23*

- Designed a comprehensive tech stack for autonomous control of the drone using external and onboard camera and various **sensors**
- Successfully developed the software and hardware stack of a **swarm** consisting of nano drones and a stable **surveillance** drone
- Enabled autonomous stabilisation control for the drone utilising **onboard computation** using the data from the magnetometer
- Worked on integrating the **Laser sensor** and Optical flow sensor(**PMW 3901**), creating custom libraries for the flight controller

Fintrac Global, London | SDE Associate | *Mentor: Debasis Chakraborty* *Dec'22-Mar'23*

- Developed the brand **website** using **Next.js**, leveraging server-side rendering, efficient **routing** system and state management
- Integrated an inbuilt form, implementing **server-side validation** and processing to ensure secure handling of the user form data
- Extensively researched about **Foreign Exchange** concepts to develop a Currency Conversion API using **Django** and **MongoDB**
- Researched device ID verification, for utilizing **next.Auth** for frontend implementation and leveraged Django for backend **APIs**
- Developed a comprehensive overview for **suspending devices** involved in fraudulent activities and thus requiring re-registration
- Received a permanent membership of the brand and also a full time **Pre Placement Offer** in the London Head Office.

KEY PROJECTS

Swarm Robots | Robofest - Gujarat 3.0 | *Mentor: Prof. Ketan Rajawat* *Jan'23 - Present*

- Designed and built a swarm of 5 stable drones deployed for **search and pickup** and pre-planned autonomous **formation flying**
- Created simulations for our drones using **Ardupilot SITL** (Software in the loop) environment and **MAVROS** Communication
- Used ROS to establish a **master-slave model** for connecting the drones over **LAN** and using **Mavlink Protocol** for controls

Pluto Drone Swarm Challenge | Inter - IIT Tech Meet 11.0 - IIT Kanpur *Dec'22 - Feb'23*

- Designed and implemented an agile **MPC Controller** derived from functionality of the **PID controller**, for Pluto nano drones
- Worked on Vision based **detection** and **positioning** of the drone using **ArUco** Marker and **Kalman-Filtered** the data received
- Created a wrapper for drone communication via **Multi-Wii Serial Protocol** and used **multi-threading** for parallel controls

Mathematical tools for Data Science | Stamatics Society IITK *Apr'22 - Jul'22*

- Learnt and implemented **mathematical tools** used in **data extraction**, **analysis**, and **calculation** in the field of Data Science
- Learnt various libraries of Python like **NumPy**, **Pandas**, **Matplotlib** and **Scikit learn** and created various small and big applications for examining data and used concepts of **Machine Learning** to extract the data and represent it in a useful form

Python and its common uses | Association for Computing Activities(ACA) IITK *Jun'22 - Jul'22*

- Worked diligently for mastering the fundamentals of Python including data types, control flow statements, functions and classes
- Delved into **web scraping** using Beautiful Soup 4, gaining skills in extracting and parsing data from HTML and XML documents
- Further working to build IITK Professor search application using **Django** to develop a research information portal for the students

TECHNICAL SKILLS

Programming languages	Robotics	Utilities	Libraries and Frameworks
Assembly, C/C++, Python, MATLAB, HTML, Java, CSS, Javascript	ROS, Gazebo, OpenCV, Arduino, RViZ, Simulink, Ardupilot, BOOST	Markdown, Git/Github, MS office, SQL, Fusion 360, Bash, Linux, AutoCAD inventor	NumPy, Pandas, Node.js, React.js, Next.js, Django, requests, BS4, sklearn

POSITIONS OF RESPONSIBILITY

Team Head | Team Aerial Robotics - IIT Kanpur *Mar'23 - Present*

- Responsible for developing and maintaining the **software** and **hardware** stack for our **custom-built fleet** of aerial robots
- Responsible for the maintenance of the team website built using **Astro** and **mentoring** junior members developing the website
- Recruited and mentoring a team of **20 members** in domains like simulation, computer vision and robot autonomy

Secretary, Web | Games and Sports Council - IIT Kanpur *Jun'22 - May'23*

- Maintained and improved the **council website**, improving the CSS and overall UI/UX of the website by adding dynamic elements

RELEVANT COURSEWORK

Fundamentals of Computing(A)	Supervised Machine Learning(*)	Introduction to Electronics
Linear Algebra & Ordinary Differential Equations	Python for Data Science, AI and Development (*)	Real Analysis